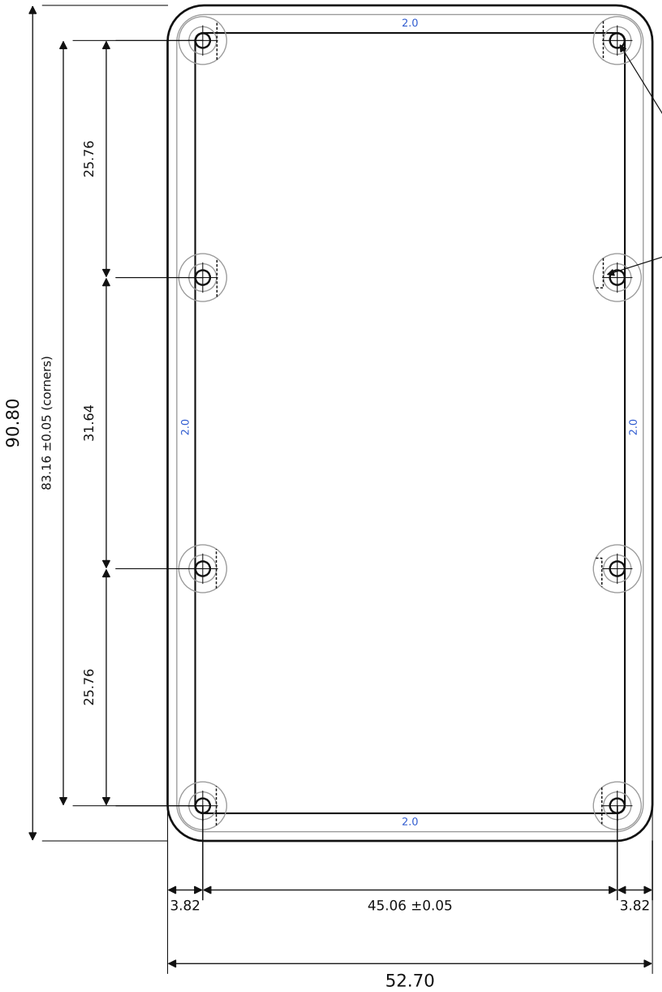
BACK FACE (outer) SCALE 1.5:1



8× BOSS FLAT (HIDDEN — CAVITY SIDE): EACH Ø5.20 BOSS CUT BACK 0.25–0.75 CLEAR OF THE PART IT WOULD FOUL (SC1 SC2 SC3 SC4), FULL DEPTH; FLATS 1.47–1.68 FROM THE HOLE AXIS (THREAD KEEP-OUT r1.30 UNTOUCHED). MODELED — STEP GOVERNS.

POS	Ø0.10 (M)	A	B

8× M2 THREAD, THRU

Ø1.6 TAP DRILL • TAP FROM BACK FACE

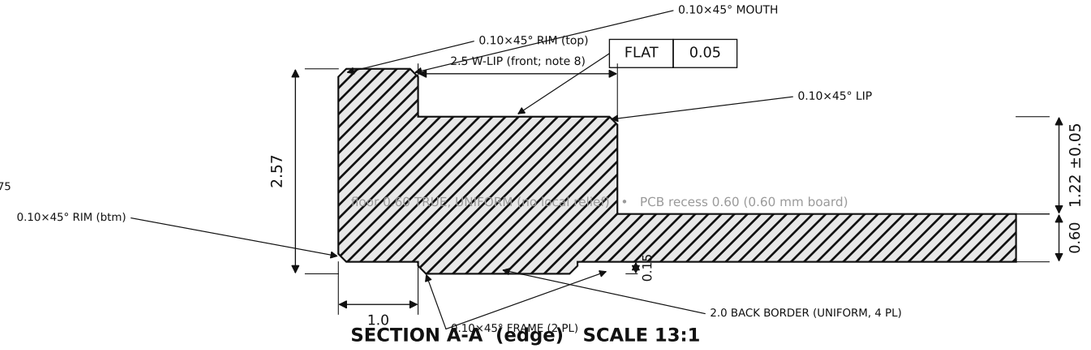
Ø3 SPOTFACE, BACK (depth per Detail B)

C3 = hole-pattern pitch ±0.05   C4 = M2 thread

SECTION A-A → edge profile   DETAIL B → corner boss

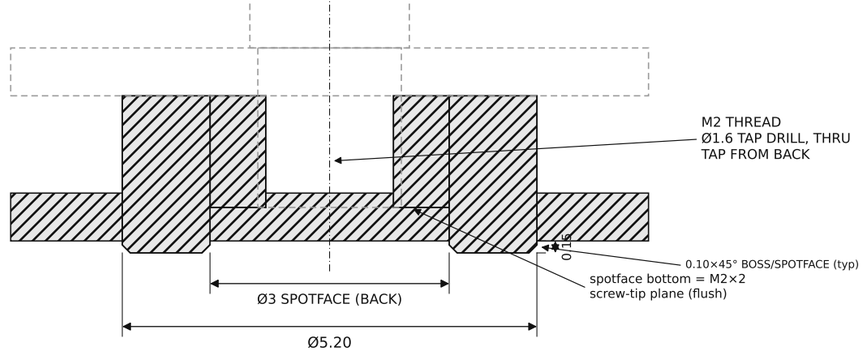
REAR ART — NONE ON THIS VARIANT

THE 0.60 FLOOR CANNOT TAKE THE FIN VALLEYS (0.60 DEEP) OR THE MEDALLION FLOORS (~0.55-0.75) — PHYSICS-FORCED OFF IN fit_rules.VARIANTS, NOT STYLING. BACK FACE = PLAIN RECESSED ART FIELD INSIDE THE 2.0 BORDER + 8 BOSS ANNULI, NOTHING ELSE.



SECTION A-A (edge) SCALE 13:1

C1 = cavity depth   C2 = PCB-rest-plane flatness   (back face down; PCB drops in from top)



DETAIL B (corner boss) 13:1

CRITICAL DIMENSIONS — C1 & C3 TOLERANCED ±0.05 (MARKED ON VIEWS); ALL OTHER FEATURES PER ISO 2768-1 (MEDIUM)

C1	Cavity depth (boss-top plane → cavity floor)	1.22 ±0.05
C2	PCB-rest plane flatness (lip + 8 bosses)	FLAT 0.05
C3	8× mounting holes (x 45.06; y rows 2.87/28.63/60.27/86.03)	±0.05 (linear)
C4	Mounting holes — tapped, thru, from back	M2 / Ø1.6 drill

NOTES

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5. [Ti Gr5 (thinned floor; the 0.60 is the shop-minimum conversation)]
2. FINISH: BEAD-BLAST MATTE, ALL OVER. NO MARKING SERVICE. NO REAR ART — THE 0.60 FLOOR CANNOT TAKE THE FIN VALLEYS (0.60 DEEP) OR THE MEDALLION FLOORS (~0.55-0.75); BOTH PHYSICS-FORCED OFF (fit_rules.VARIANTS).
3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ±0.05. DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE.
4. BRACE REGISTRATION: THE RESIN H-BRACE REGISTERS TO THIS SHELL BY FITMENT - ITS 4 OUTBOARD RAILS + THE COMPONENT POCKETS + THE BOARD PRESS-FIT. NO LOCATOR PILLARS; THE CAVITY FLOOR IS A FULL 0.60 EVERYWHERE.
5. EDGE BREAKS - ALL 0.10x45°, FELT NOT SEEN, MODELED. CALLED OUT ON SEC A-A / DETAIL B: RIM = outer rim (top & bottom); MOUTH = recess mouth (around PCB); LIP = inner (cavity-side) lip edge; FRAME = proud back-frame bottom edges; BOSS = boss + spotface bottom edges (Detail B). BREAK ALL OTHER EXPOSED EDGES 0.10x45°. CONCAVE JUNCTION CORNERS (BOSS
6. FLOOR IS A TRUE 0.60, UNIFORM ACROSS THE CAVITY, BELOW THE ~1.0 Ti MIN-WALL GUIDANCE - THE 0.60 IS THE SHOP-MINIMUM CONVERSATION (fit_rules.VARIANTS); CONFIRM THE SHOP HOLDS IT BEFORE QUOTING.
   CAVITY 1.22 ±0.05 -> 1.17 WORST-CASE, MINUS THIN CAP 1.00 MAX (PROVISIONAL — part_heights.SUPERCAP_H_THIN) = 0.17 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD.
7. NO U7 RELIEF POCKET. U7 IS THE 0.90 mm MB85RC512TY DFN-8, WELL UNDER THE 1.22 CAVITY, SO THE FLOOR IS UNIFORM 0.60. (A POCKET EXISTED FOR A 1.75 SOIC-8 THE v4 BOARD DOES NOT CARRY.)
8. FRONT SUPPORT LIP IS ASYMMETRIC (SECTION A-A): WEST 2.5 / NORTH 2.0 / SOUTH 2.0 FOR PCB RIGIDITY; EAST 1.0 CLEARING THE JP1/TP1 PADS, THE NFC COIL (A GROUNDED LIP WOULD DETUNE IT), AND C7 (x49.55, THE ONE EAST-EDGE PART LEFT AFTER v4 REMOVED THE Q1/U4/R7/R9 CLAMP CLUSTER + D9/D10/D11); WIDENING TO 2.5 ONLY AT THE SOUTH END (y0-10, CLEAR C
9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE.
10. PCB RECESS = 0.60 DEEP (RECEIVES THE 0.60 mm BOARD; SLIP FIT, NOT A PRESS FIT). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC.
11. PART = BACK-SHELL ONLY. PCB, RESIN BRACE, AND 8× M2×2 BRASS SLOTTED-CHEESE SCREWS (HEAD Ø3.8) SUPPLIED SEPARATELY.

GENERAL TOLERANCES (UNLESS OTHERWISE NOTED, mm)	
LINEAR & HOLE-POSITION DIMS	**ISO 2768-m**
CAVITY DEPTH C1 (SECTION A-A)	**1.22 ±0.05**
HOLE-PATTERN PITCH C3 (8 HOLES)	**x 45.06 / y 25.76-31.64-25.76 ±0.05**
PCB-REST FLATNESS C2	**FLAT 0.05**

SOLAR-GLOW DRH — Ti BACK-SHELL LITE (0.60 FLOOR, THINNED)	
DWG solar-glow-drh-shell-lite-Ti	REV lite
MATERIAL Ti Gr5 (TC4)	FINISH BEAD-BLAST MATTE
UNITS mm	SCALE AS NOTED
BBOX 52.70 x 90.80 x 2.57 3rd-angle	SHEET 1/1